深圳市永达创力电子有限公司

Shenzhen yongda chuangli electronic co., LTD

SPECIFICATION

LCD MODULE

CL035C013

REVISION RECORD

DESIGN	CHECK	REVIEW
VERSION	DATE	CONTENTS
A	2016-04-29	First Release
B	2018-11-22	Function Description

CUSTOMER

Customer company: Date:

Customer signature: Date:

CONTENTS

- GENERAL INFORMATION
- ABSOLUTE MAXIMUM RATINGS
- ELECTRICAL CHARACTERISTICS
- TIMING OF POWER SUPPLY
- BACKLIGHT CHARACTERISTICS
- EXTERNAL DIMENSIONS
- Module Function Description
- INTERFACE SIGNALS
- ELECTRO-OPTICAL CHARACTERISTICS
- QUALITY LEVEL
- PACKAGING DRAWING
- PRECAUTIONS FOR USE OF LCD MODULES

GENERAL INFORMATION

Item	Contents	Unit
Driver element	a-Si 3.5 TFT active matrix	--
Viewing direction	12 O'CLOCK	O' Clock
LCM OUTLINE DIMEMSIONS	54.48(W) x 84.71(H) x 2.2(T)	mm
Active area (W×H)	48.96(H) × 73.44(V)MM	mm
Number of Dots	320RGB(H)×480(V)	Pixel
Driver IC	ILI9488	--
Colors	262K	--
Weight	TBD	g
Backlight Type	LED	--
Interface Type	RGB or 8/9/16/18 Parallel Interface	--
Input voltage	2.8	V

ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Min	Max	Unit
Supply voltage for analog	V_{DD}	-0.3	3.3	V
Input voltage	V_{IN}	-0.3	V_{DD}+0.3	V
Operating temperature	T_{OP}	-20	70	℃
Storage temperature	T_{ST}	-30	80	℃
Humidity	RH		90%(Max60℃)	RH

ELECTRICAL CHARACTERISTICS

DC CHARACTERISTICS

Parameter	Symbol	Min	Typ	Max	Unit
Supply voltage for analog	V_{DD}	2.3	2.8	3.3	V
Input Current	Idd	—	-	-	mA
Supply voltage for I/O circuit	IOVCC	1.65	1.8	3.3	V
Input voltage ' H' level	V_{IH}	0.7 IOVCC	—	—	V
Input voltage ' L' level	V_{IL}	—	—	0.3 IOVCC	V
Output voltage ' H' level	V_{OH}	0.8 IOVCC	—	—	V
Output voltage ' L' level	V_{OL}	—	—	0.2 IOVCC	V

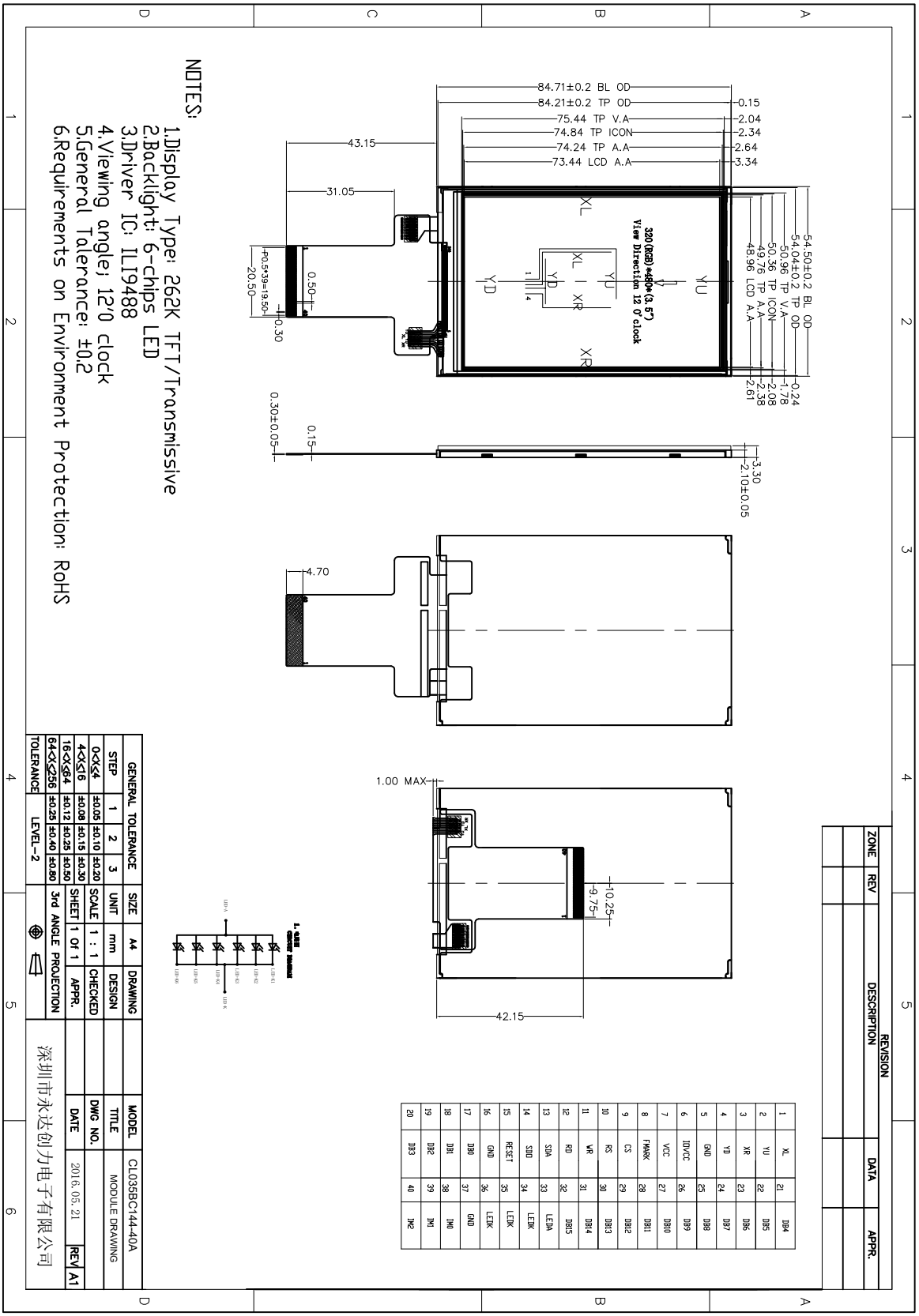
TIMING OF POWER SUPPLY

PLEASE REFER TO THE DRIVER IC SPECIFICATION.

BACKLIGHT CHARACTERISTICS

Item	Symbol	Min	Typ	Max	Unit	Condition
Forward voltage	Vf	2.9	3.2	3.3	V	If=120 mA
Luminance	L_V	6000	--	--	cd/m²	
Number of LED	-	6			Piece	-
Connection mode	p	Parallel			-	-

EXTERNAL DIMENSIONS



Module Function Description

Pin Descriptions

PIN No.	Symbol	Description
1	X(L)	Touch panel control pin (触摸屏控制脚)
2	Y(U)	Touch panel control pin (触摸屏控制脚)
3	X(R)	Touch panel control pin (触摸屏控制脚)
4	Y(D)	Touch panel control pin (触摸屏控制脚)
5	GND	Ground （接地脚）
6	IOVCC	Power supply for LCM（2.8V-3.3V） （屏供电脚）
7	VCC	Power supply for LCM（2.8V-3.3V） （屏供电脚）
8	FMARK	Tearing effect output pin to synchronize MPU to frame writing, activated by S/W command. When this pin is not activated, this pin is low. If not used, open this pin. (帧同信号,不用时悬空)
9	CS/SPI CS	Chip select pin (“Low” enable) （屏驱动芯片片选脚，低电平有效）
10	RS/SPI A0(4线)	This pin is used to select “Data or Command” in the parallel interface or serial data interface. (用于并口或者串口) Parallel(并口): When RS= ’1’, data is selected.(选择数据) When RS= ’0’, command is selected.(选择寄存器) Serial(串口): 4-line system (D/CX): Serves as command or parameter select. *Fix to*IOVCC *level when not in use.* （并口的寄存器/数据选择控制脚或者4线串口的寄存器/数据选择,不用时接IOVCC）

		If not used, this pin should be connected to IOVCC or GND. (不用时接 IOVCC 或者接地)
11	WR/SPI SCL	Serves as a write signal and writes data at the rising edge. This pin is used serial interface clock in 3-wire 9-bit / 4-wire 8-bit serial data interface. (3线串口或者4线串口的时钟信号) （并口的写控制脚或者3线/4线串口的时）
12	RD	Serves as a read signal and MCU read data at the rising edge. *Fix to* IOVCC *level when not in use.* （并口的读控制脚,不用时接 IOVCC）
13	SPI SDI/SDA	Serial input signal. The data is applied on the rising edge of the SCL signal. If not used, fix this pin at IOVCC or GND (串口数据输入信号,不用时接 IOVCC 或者接地)
14	SPI SDO	Serial output signal. The data is outputted on the falling edge of the SCL signal. If not used, open this pin (串口数据输出信号, 不用时悬空)
15	RESET	LCM Reset pin Signal is active low. （屏复位脚,低电平复位）
16	GND	Ground （接地脚）
17-24	DB0-DB7	Data bus *Fix to* GND *level when not in use* (低 8 位数据线,不用时接地)
25-32	DB8-DB15	Data bus *Fix to* GND *level when not in use* (高 8 位数据线,不用时接地)
33	A	Anode of Backlight (3.0V-3.4V Typical:3.2V) （背光正极供电脚,电压范围:3.0-3.4V,典型值:3.2V）
34-36	K	Cathode of Backlight （背光负极供电脚）
37	GND	Ground （接地脚）

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>38</td><td>IM0</td><td rowspan="3">Select the MCU interface mode
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>Interface mode</th></tr>
<tr><td>0</td><td>0</td><td>1</td><td>DBI Type B 9-bit bus DB [8:0]</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>DBI Type B 16-bit bus DB [15:0]</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>DBI Type B 8-bit bus DB [7:0]</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>DBI Type C Option 1(3-line SPI)</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>DBI Type C Option 1(4-line SPI)</td></tr>
</table></td></tr>
<tr><td>39</td><td>IM1</td></tr>
<tr><td>40</td><td>IM2</td></tr>
</table>

APPLICATION CIRCUIT

Please consult our technical department for detail information.

INITIAL CODE

Please consult our technical department for detail information.

ELECTRO-OPTICAL CHARACTERISTICS

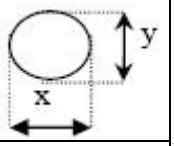
Item		Symbol	Condition	Min	Typ	Max	Unit	Remark	Note
Response time		Tr+Tf	θ =0° ☛=0° Ta=25℃	–	20	30	ms	FIG 1.	4
Contrast ratio		Cr		–	400	500	–	FIG 2.	1
Luminance uniformity		δ WHITE		–	–	–	%	FIG 2.	3
Surface Luminance		Lv		300	–	–	cd/m²	FIG 2.	2
Viewing angle range		CR>10	☛ 3	60	70	–	deg	FIG 3.	6
			☛ 9	60	65	–	deg	FIG 3.	
			☛ 12	60	70	–	deg	FIG 3.	
			☛ 6	50	65	–	deg	FIG 3.	
CIE(x,y) chromaticity	Red	x	θ =0° ☛=0° Ta=25℃	0.586	0.636	0.686		FIG 2.	5
		y		0.273	0.323	0.373			
	Green	x		0.252	0.277	0.297			
		y		0.529	0.549	0.569			
	Blue	x		0.122	0.142	0.162			
		y		0.102	0.122	0.142			
	White	x		0.283	0.303	0.323			
		y		0.305	0.325	0.345			

4.Standards of inspection items

4.1Major Defect

Item No	Items to be inspected	Inspection Standard	Classification of defects
4.1.1	All functional defects	1.No display 2.Display abnormally 3.Missing vertical, horizontal segment 4.Short circuit 5. Back-light no lighting, flickering and abnormal lighting.	Major
4.1.2	Missing	Missing component	
4.1.3	Outline dimension	Overall outline dimension beyond the drawing is not allowed.	
4.1.4	linearity	No more than 1.5%	

4.2 Cosmetic Defect

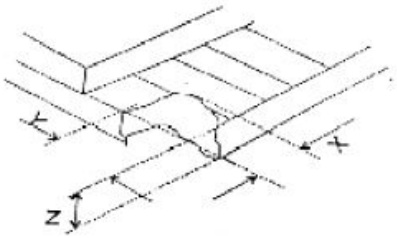
Item No	Items to be inspected	Inspection Standard				Classification of defects
4.21	**Clear Spots** Black and white Spot defect Pinhole, Foreign Particle, polarizer Dirt	For dark/white spot, size Φ is defined as $\Phi=\frac{(x+y)}{2}$ 				Minor
		1				
		Zone / Size(mm)	Acceptable Qty			
			A	B	C	
		Φ≤0.15	Ignore		Ignore	
		0.15＜Φ≤0.20	2			
		0.20＜Φ≤0.30	1			
		Φ＞0.30	0			
	Clear Spots TP Dirt	2				Minor
		Zone / Size(mm)	Acceptable Qty			
			A	B	C	
		Φ≤0.15	Ignore		Ignore	
		0.15＜Φ≤0.20	2			
		0.20＜Φ≤0.30	1			
		Φ＞0.30	0			
	Dim Spots Circle shaped and dim edged defects	3				Minor
		Zone / Size(mm)	Acceptable Qty			
			A	B	C	
		Φ≤0.2	Ignore		Ignore	
		0.20＜Φ≤0.40	2			
		0.40＜Φ≤0.60	1			

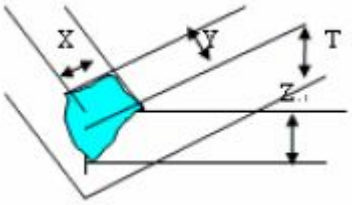
		Φ＞0.60	0		

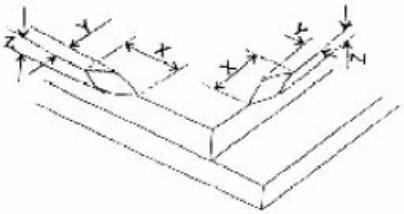
2 Cosmetic Defect

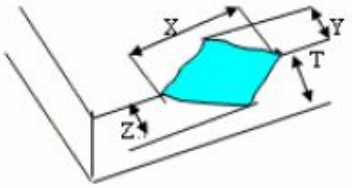
Item No	Items to be inspected	Inspection Standard					Classification of defects
4.2.2	**Line defect** Black line, White line, Foreign material on polarizer	Size(mm)		Acceptable Qty			Minor
		L(Length)	W(Width)	Zone			
				A	B	C	
		Ignore	W≤0.01	Ignore		Ignore	
		L≤3.0	0.01＜W≤0.03	2			
		L≤3.0	0.03＜W≤0.05	1			
			W＞0.05	0			
	Foreign material on TP film	The line can be seen after mobile phone in the operating condition:					Minor
		Size(mm)		Acceptable Qty			
		L(Length)	W(Width)	Zone			
				A	B	C	
		Ignore	W≤0.03	Ignore		Ignore	
		L≤5.0	0.03＜W≤0.05	3			
			W＞0.05	0			
4.2.3	**Dim line defect** Polarizer scratch TP film scratch	If the scratch can be seen after mobile phone cover assembling or in the operating condition, judge by the line defect of 4.2.2. If the scratch can be seen only in non-operating condition or some special angle, judge by the following.					Minor
		Size(mm)		Acceptable Qty			
		L(Length)	W(Width)	Zone			
				A	B	C	
		Ignore	W≤0.03	Ignore		Ignore	
		5.0＜L≤10.0	0.03＜W≤0.05	2			
		L≤5.0	0.05＜W≤0.08	1			
			W＞0.08	0			
4.2.4	Polarize Air bubble	Air bubbles between glass & polarizer					Minor
		Zone / Size(mm)		Acceptable Qty			
				A	B	C	
		Φ≤0.25		Ignore		Ignore	
		0.25＜Φ≤0.5		2			
		Φ＞0.50		0			

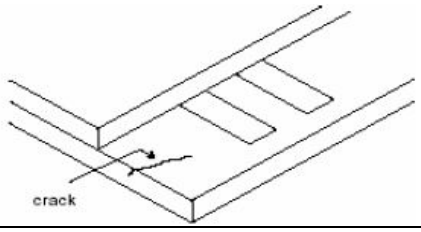
Item No	Items to be inspected	Inspection Standard	Classification of defects

<table>
<tr>
<td>4.35</td>
<td>Glass defect</td>
<td>
(i) Chips on corner
A:LCD Glass defect


<table>
<tr><td>X(mm)</td><td>Y(mm)</td><td>Z(mm)</td></tr>
<tr><td>≤2.0</td><td>≤S</td><td>Disregard</td></tr>
</table>
Notes: S=contact pad length
Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal.
B:TP Glass defect


<table>
<tr><td>X(mm)</td><td>Y(mm)</td><td>Z(mm)</td></tr>
<tr><td>≤3.0</td><td>≤3.0</td><td>Disregard</td></tr>
</table>
(ii)Usual surface cracks
A:LCD Glass defect


<table>
<tr><td>X(mm)</td><td>Y(mm)</td><td>Z(mm)</td></tr>
<tr><td>≤3.0</td><td><Inner border line of the seal</td><td>Disregard</td></tr>
</table>
B:TP Glass defect


<table>
<tr><td>X(mm)</td><td>Y(mm)</td><td>Z(mm)</td></tr>
<tr><td>≤6.0</td><td>＜2.0</td><td>Disregard</td></tr>
</table>
(iii) Crack
Cracks tend to break are not allowed.


</td>
<td>Minor</td>
</tr>
</table>